

Lecture 2: Heuristics and Biases

Mental Shortcuts and Their Systematic Consequences

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ECON 230 · Introduction to Behavioural Economics · Winter 2027

Which Kills More People?

Quick Quiz: Which kills more Americans/year? (A or B)

A	B
1. Shark attacks	Falling out of bed
2. Deer (car collisions)	Lightning strikes
3. Tornadoes	Bee stings
4. Plane crashes	Car crashes (per km)

Answers (CDC):

- Bed falls **450/yr** vs Sharks **6/yr**
- Deer **130/yr** vs Lightning **50/yr**
- Bee stings **62/yr** vs Tornadoes **70/yr**
- Cars **1.37/100M km** vs Planes **0.07**

Why Wrong?

Sharks, planes, tornadoes: **vivid and memorable**. Bed falls: quiet, no headlines. This is the **Availability Heuristic**.

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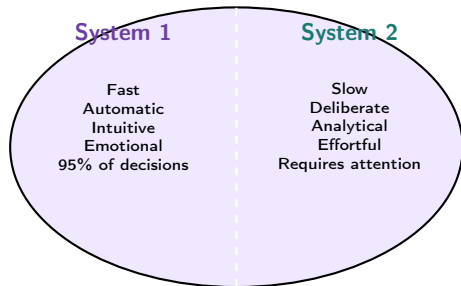
Five Heuristics and Biases Today

1. **Dual Process Theory** — System 1 vs System 2
2. **Representativeness** — Judge by resemblance (Linda Problem, Base Rate Neglect)
3. **Availability** — Judge by ease of recall (risks, markets)
4. **Anchoring** — Adjust from a starting point (insufficiently)
5. **Confirmation Bias** — Seek confirming evidence
6. **Overconfidence** — Overestimate accuracy and ability

Key Message

These are not random errors. They are **systematic, predictable, and stable across populations**. That is what makes them economically important.

Dual Process Theory



System 1 Examples

Reading a face, driving a familiar route, 2×2 , catching a ball, speaking your native language.

System 2 Examples

Solving 17×24 , navigating a new city, writing an economics essay, following a complex legal argument.

Kahneman's Key Insight

System 1 is “lazy System 2’s” servant.
System 2 endorses System 1’s output without checking — most of the time.

Decision Fatigue — System 2 Gets Tired

The Israeli Parole Board (Danziger et al., 2011)

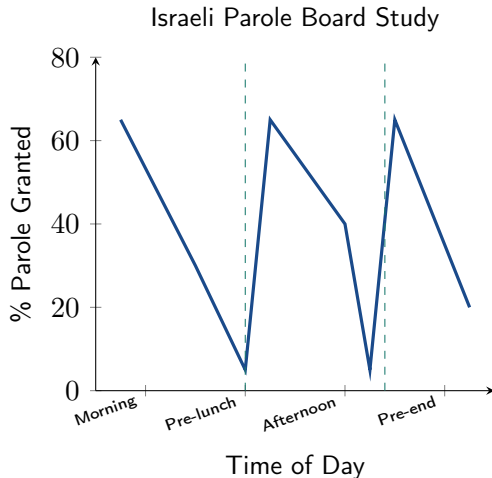
Analysed 1,112 parole board decisions across a full day.

- **Start of session:** 65% approval rate.
- **Just before food break:** Near 0% approval.
- **After food break:** Back to 65%.

The pattern repeated three times per day.

Interpretation

When System 2 is depleted by decision-making, it defaults to the **safe, status-quo option** (deny parole). The prisoner's fate depended on *when* the hearing was scheduled.



The Representativeness Heuristic

Definition (Kahneman & Tversky, 1972)

We judge the probability that object X belongs to category Y by how much X *resembles* our mental prototype of Y — **ignoring base rates and sample size**.

Classic Example

Tom is a quiet, detail-oriented person who loves puzzles and sci-fi. Is Tom more likely to be a librarian or a truck driver?

The Answer

Truck driver — by far. There are ≈ 3.5 million truck drivers in the US vs $\approx 170,000$ librarians. The base rate alone predicts truck driver, even if Tom fits the librarian stereotype perfectly.

Representativeness ignores the **prior probability** (base rate) and focuses on the **likelihood** of the description.

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Linda's Description

"Linda is 31 years old, single, outspoken, and very bright. She majored in philosophy. As a student, she was deeply concerned with discrimination and social justice, and participated in anti-nuclear demonstrations."

Rank by Probability (write your answer)

- A. Linda is a bank teller.
- B. Linda is a bank teller and is active in the feminist movement.

The Answer: $A \geq B$. *Always*. By the laws of probability.

$P(\text{bank teller} \cap \text{feminist}) \leq P(\text{bank teller})$. Yet 85% of subjects chose B (Tversky & Kahneman, 1983). Replicated in doctors, statisticians, and Nobel Prize winners.

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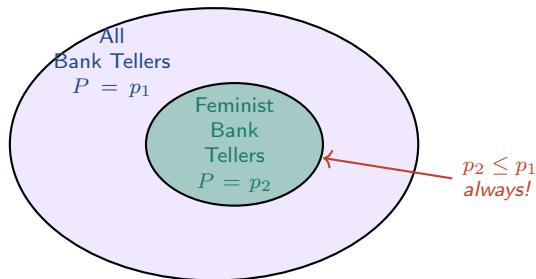
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The Conjunction Fallacy — A Venn Diagram



The Conjunction Rule

$P(A \cap B) \leq P(A)$ for any events A, B .

Adding conditions can **only reduce or maintain** probability. It can *never* increase it.

Why People Violate This

Linda's description **resembles** a feminist bank teller more than a generic bank teller. Representativeness overrides probability theory.

Economic Implication

Investors believe stories like “tech stock rises because AI boom AND rate cuts” is more likely than just “tech stock rises” — leading to overconfidence in complex narratives.

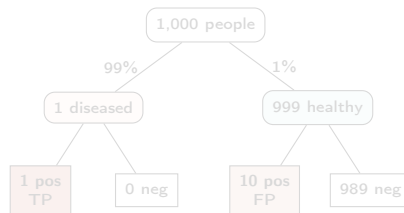
Base Rate Neglect — Medical Diagnosis

The Scenario

- Disease affects **1 in 1,000** people.
- Test is **99% accurate** (1% false positive rate).
- You test positive. What is the probability you have the disease?

Most People Say: $\approx 99\%$. Correct Answer: $\approx 9\%$

- 1,000 people tested: 1 true positive (tests positive), 10 false positives.
- $P(\text{disease}|+) = \frac{1}{1 + 10} \approx 9\%$



Of 11 positives: only 1 is truly diseased!

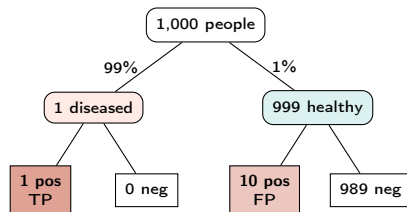
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The Gambler's Fallacy

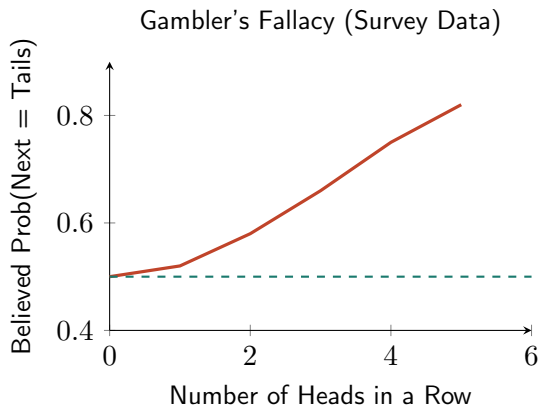
Definition

After a streak (e.g., 5 heads in a row), people believe the opposite is now “due.” The random process is expected to **self-correct**.

Why It Happens: HHHHH does not *look* random; HTTHHT does. Representativeness drives expectations.

Economic Consequences

- Casino players bet against streaks at roulette.
- Lottery players avoid “recent” winning numbers.
- Loan officers less likely to approve after approving loan $n - 1$ (Rao et al., 2019).



The Availability Heuristic

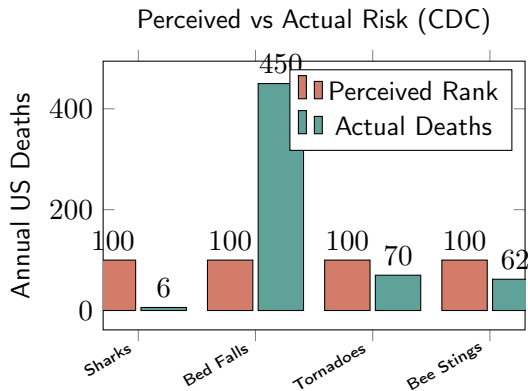
Definition (Tversky & Kahneman, 1973)

We judge the frequency or probability of events by how **easily examples come to mind**. Vivid, recent, dramatic events are more available — and therefore judged more likely.

Classic Demonstration

“Are there more English words that start with K, or words with K as the 3rd letter?”

- Most say: *start with K*
- Correct: *3rd letter* (by a factor of $\approx 3\times$)
- It's just much easier to retrieve K-first words.



Availability Cascade

Kuran & Sunstein (1999)

An **availability cascade**: a self-reinforcing cycle where:

1. A minor risk event occurs.
2. Media covers it intensively — it becomes highly available.
3. Public perceives the risk as much larger.
4. More media attention follows.
5. Policy responds to the inflated public fear.

Alar Pesticide Scare (1989)

CBS 60 Minutes reported Alar (used on apples) as “the most potent cancer-causing agent in the food supply.” Zero documented deaths. Yet apple sales collapsed 25% in 3 months. Schools banned apples. Cost to industry: \$375M.

Contrast: Radon in Homes

Radon kills $\approx 21,000$ Americans per year — 3rd leading cause of lung cancer. Almost no media coverage. Almost no public concern. Low availability = low perceived risk.

The Anchoring Effect

Tversky & Kahneman (1974)

When making quantitative judgments, people start from an initial value (the **anchor**) and adjust — but they adjust **insufficiently**. The final estimate is biased toward the anchor, even if it is *clearly arbitrary*.

The Spinning Wheel Experiment

Tversky & Kahneman had subjects spin a “rigged” wheel that landed on either 10 or 65. They were then asked: “What % of African nations are in the UN?”

- Wheel landed on 10: mean estimate = **25%**
- Wheel landed on 65: mean estimate = **45%**
- The random wheel number shifted answers by **20 percentage points**.

Why Do Irrelevant Numbers Matter?

Even when people know the anchor is random, they cannot fully disengage from it. The anchor activates nearby numeric concepts in memory, biasing retrieval.

Group A — Left Side

1. Is a bottle of 1998 Bordeaux wine worth **more or less than \$10**?
2. Write your **best estimate** of its value.

Group B — Right Side

1. Is a bottle of 1998 Bordeaux wine worth **more or less than \$400**?
2. Write your **best estimate** of its value.

Expected Results (Ariely, Loewenstein & Prelec, 2003)

- Group A mean: $\approx$ \$20–35
- Group B mean: $\approx$ \$85–130
- Neither group can verify the correct price — yet the anchor dominates.
- Ariely et al. called this “**arbitrary coherence**”: once anchored, people are internally consistent but externally anchored.

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Anchoring in Real Markets

Real Estate (Northcraft & Neale, 1987)

Experienced real estate agents were shown the same property with different listing prices. Their estimated values differed by up to **11%**. The listing price anchored even expert valuers.

Salary Negotiation

Galinsky & Mussweiler (2001): the first salary offer anchors the final settlement. Making a first offer first — even a high one — predicts 10–15% higher final salary.

Retail Anchoring

“Compare at \$200 — Our price: \$79” exploits anchoring. The fictitious \$200 anchors perceived value, making \$79 seem like a bargain, regardless of actual market value.

Legal Sentencing (Englich et al., 2006)

German judges were asked to roll a die before sentencing a shoplifter.

- Die rolled 3: recommended sentence = **5 months**
- Die rolled 9: recommended sentence = **8 months**

A random number shifted prison sentences by 50%. Judges knew the die was random.

Definition

We search for, interpret, favour, and recall information in a way that **confirms our pre-existing beliefs**. Disconfirming evidence receives less attention and is more easily dismissed.

Wason (1960) Selection Task

Cards: each has a number on one side and a letter on the other. You see: **E, K, 4, 7**. Rule: "If a card has E on one side, it has 4 on the other." Which cards must you turn over?

Correct Answer: E and 7

Most say E and 4. They seek *confirming* cases (E and 4) and ignore the falsifying case (7 with E on the back would violate the rule). Only 10% solve this correctly.

In Financial Markets

Investors interpret ambiguous earnings news as confirming their existing position.

- Bull: "Inflation fell — great for growth!"
- Bear: "Inflation fell — recession signal!"

Lord, Ross & Lepper (1979): subjects who read *the same* mixed evidence study both concluded it *strengthened* their prior views.

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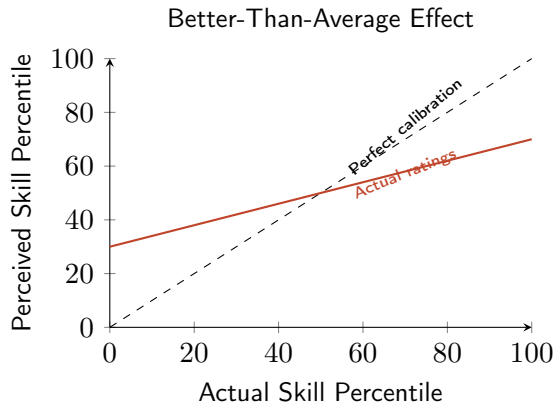
Three Types of Overconfidence

Moore & Healy (2008) Taxonomy:

1. **Overprecision:** Your 90% confidence intervals are too narrow.
2. **Overplacement (Better-Than-Average):** You think you perform above your peer group.
3. **Overestimation:** You overestimate your absolute ability or performance.

Key Data

- **93%** of US drivers: above average (Svenson, 1981)
- **94%** of professors: above-average teachers (Cross, 1977)
- **84%** of French drivers: above average
- Finance professors: 80% CI captures actual S&P return only **33%** of the time (Graham & Harvey, 2003)



The Dunning-Kruger Effect

Kruger & Dunning (1999)

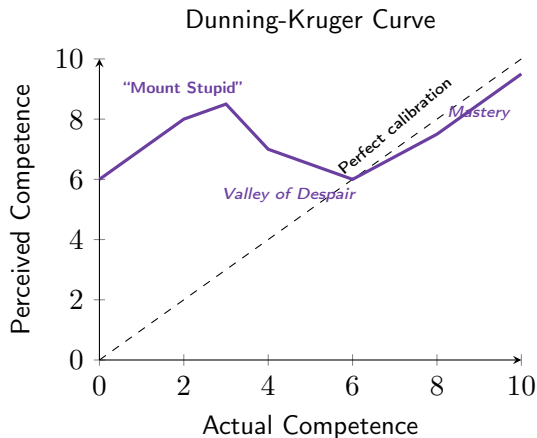
Low-competence individuals most severely overestimate their ability. High-competence individuals slightly underestimate.

"The trouble with the world is that the stupid are cocksure and the intelligent are full of doubt."

— Bertrand Russell

Application

Entry-level managers who don't know what they don't know. New investors who made money in a bull market. First-year students who haven't encountered hard questions yet.

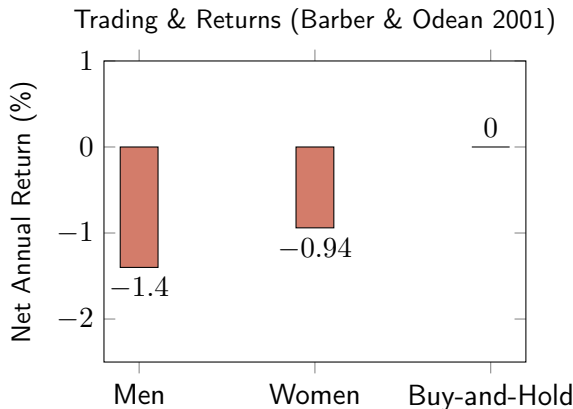


Overconfidence and Financial Markets

Barber & Odean (2001) — “Boys Will Be Boys”

- Men trade **45% more**; net returns **1.4% lower** than buy-and-hold.
- Women trade less; net returns **0.94% lower**. Both lose to inaction.
- Overconfidence $\Rightarrow$ excess trading $\Rightarrow$ destroyed returns.

The Planning Fallacy: Sydney Opera House: \$7M $\rightarrow$ \$102M. Denver Airport: \$1.7B over. Montreal Olympics: debt paid in **2006**.



How Heuristics Interact

Heuristic	Core Mechanism	Key Failure
Representativeness	Judge by resemblance	Base rate neglect; conjunction fallacy
Availability	Judge by ease of recall	Overweight vivid/recent; media bias
Anchoring	Adjust from starting point	Insufficient adjustment; arbitrary anchors
Confirmation	Seek confirming evidence	Polarisation; bad decisions
Overconfidence	Overestimate own accuracy	Excess trading; planning failures

They Amplify Each Other

Overconfidence reduces the search for disconfirming evidence (confirmation bias). Representativeness makes vivid narratives more available. Anchoring interacts with confirmation bias to lock in initial assessments.

The Problem (Graber et al., 2005)

Diagnostic error rate in medicine: $\approx 15\%$ of cases. In Canada: $\approx 50,000$ preventable patient harms per year linked to diagnostic errors (CPSI, 2004).

Which Heuristic?

- **Anchoring:** First diagnosis anchors all subsequent interpretation.
- **Availability:** Recent dramatic cases shape diagnosis (“Zebra” thinking: rare disease imagined after seeing one last week).
- **Representativeness:** Patient “doesn’t fit the profile” $\Rightarrow$ miss the diagnosis.
- **Overconfidence:** “I know what this is.”
Premature closure.

Kahneman’s Solution

Structured checklists (Gawande, 2009).
“Consider the opposite” — force generation of one alternative diagnosis. Reduces availability + anchoring effects by 30–40%. Canadian hospitals adopting structured diagnostic protocols (CPSI 2018 guidelines).

Discussion Questions

- Q1. “Can you think of a major decision in your life (career, university, relationship) where one of today’s heuristics played a role? Which one?”
- Q2. “Should policymakers try to ‘debias’ people through education? Is it possible? Who decides what counts as a bias?”
- Q3. “Are heuristics always bad? Can you think of situations where using a heuristic leads to *better* decisions than deliberate analysis?”

Gigerenzer’s Counter-Argument

Gerd Gigerenzer (2008) argues that heuristics are *ecologically rational* — they are adapted to the environments in which humans evolved. “Less is more”: in uncertain environments, simple rules can outperform complex models. The debate between Kahneman and Gigerenzer is one of the richest in behavioural science.

Key Takeaways — Lecture 2

- ✓ **System 1 vs System 2:** Fast intuitive vs slow deliberate thinking. System 2 depletes; defaults to System 1.
- ✓ **Representativeness:** Judge by resemblance $\Rightarrow$ base rate neglect, conjunction fallacy, gambler's fallacy.
- ✓ **Availability:** Judge by ease of recall $\Rightarrow$ overweight vivid/media-salient risks.
- ✓ **Anchoring:** Adjust insufficiently from initial value $\Rightarrow$ real estate, salary, legal judgments, prices.
- ✓ **Confirmation bias:** Seek confirming evidence $\Rightarrow$ polarisation, financial errors.
- ✓ **Overconfidence:** Overestimate own performance $\Rightarrow$ excess trading, planning failures, 93% of drivers.
- ✓ These are **systematic and predictable** — which is what makes them economically consequential.

Next: Lecture 3: Prospect Theory — We formally model loss aversion and build the S-shaped value function that replaces expected utility theory.